Wind Turbine Tower Failure Modes under Seismic and Wind Loads

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Abstract: This paper studies the structural responses and failure modes of a 1.5-MW horizontal-axis wind turbine under strong ground motions and wind loading. Ground motions were selected and scaled to match the two design response spectra given by the seismic code, and wind loads were generated considering tropical cyclone scenarios. Nonlinear dynamic time-history analyses were conducted and structural performances under wind loads as well as short- and long-period ground motions compared. The results show that under strong wind loads the collapse of the wind turbine tower is driven by the formation of a plastic hinge at the lower section of the tower. This area is also critical when the tower is subject to most ground motions. However, some short-period earthquakes trigger the collapse of the middle and upper parts of the tower due to the increased contribution of high-order vibration modes. Although long-period ground motions tend to result in greater structural responses, short-period earthquakes may cause brittle failure modes in which the full plastic hinge develops quickly in regions of the tower with only a moderate energy dissipation capacity. Based on these results, recommendations for future turbine designs are proposed.
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Introduction

Wind energy is widely used nowadays as a renewable energy source despite some controversial issues surrounding it (Dai et al. 2015a). Worldwide wind energy capacity reached 539.29 GW by 2017 (WWEA 2018). Guidelines like IEC 61400 (IEC 2005) and GL (2010) are generally adopted for the design of wind turbine (WT) tower. The technology behind wind energy infrastructure developed in Northern Europe has been adopted in many regions around the world. However, the risk of extreme loading conditions such as tropical cyclonic winds or earthquakes is more significant in Asia (Dai et al. 2017a), and there are a number of WT reports on tower failures during extreme typhoons in China's southeast coastal area (Ishihara et al. 2005; Chen et al. 2015; Hawbecker et al. 2017). Failures of WT towers subject to ground motion are rare, but the risk should be carefully considered given the fact that it affects entire wind farms, where a considerable number of similarly designed and nonredundant WT towers are subject to the same seismic actions.

WT towers under wind and seismic actions have been studied through experimental testing, field investigations, and numerical analysis. The experimental testing of the collapse of WT towers under large dynamic excitations tends to be limited by the space and safety of the test site. Large-scale elastic testing has been performed, such as the large-scale experimental program on a decommissioned WT tower by Prowell et al. (2009, 2014), but few WT towers have been experimentally tested to failure. Additionally, some scale issues are involved in the experimental testing of WT towers in wind tunnels that may affect the accuracy of the results (Treuren 2015). Field investigation of failure tower following disasters can provide forensic information that is valuable from a practical point of view, but these are the final stage results of extreme events for which the failure process cannot be described. Numerical simulation techniques based on nonlinear finite-element (FE) analysis, if properly used, are capable of demonstrating the progressive collapse of a structure under extreme loading.

Regarding previous numerical studies on the seismic response of WT towers, Nuta et al. (2011) performed incremental dynamic analysis (IDA) based on the FE method to obtain fragility curves. Afterwards, Patil et al. (2016) carried out a detailed fragility analysis under near-fault and far-fault ground motions. More recently, Sadowski et al. (2017) compared the IDA results on a WT tower with welding imperfections under near- and far-fault earthquakes. With regard to the FE analysis of WT towers under extreme wind loads, Zhang et al. (2014) performed dynamic time-history analysis under simulated typhoon wind velocity histories to illustrate the WT tower failure modes. Dai et al. (2017b) conducted FE analysis of a typical 1.5 MW WT tower and observed different failure stages in wind-induced collapse for different wind directions. Furthermore, wind turbine towers have been investigated under multiple hazard loads. Asareh et al. (2016) combined seismic fragility analysis with operational wind loads based on FE analysis. Smith and Mahmoud (2016) evaluated the performance of wind turbines of various heights under wind, operation, and seismic loads. Mo et al. (2017) performed seismic fragility analysis considering different operating conditions. Although many studies have been conducted using nonlinear dynamic FE simulations, few of them specifically compared the failure process under strong ground motions or wind actions.

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This paper studies the nonlinear response of a typical WT tower, representative of many wind farms in the world, under wind and seismic actions. While long-term across-wind effects such as vortex-induced vibrations can lead to structural damage in WT towers, this study focuses on responses under extreme load cases, and thus only along-wind load effects are considered. The detailed FE model of the WT tower is presented first, followed by the proposed seismic and wind excitations. Dynamic analysis and nonlinear static (pushover) results of the WT response under wind and seismic loads are discussed in detail to point out the strong links between the loading frequency content, the structural vibration modes, and the leading collapse mechanisms.

Finite-Element Model and Modal Analysis

FE Modeling

A typical 1.5-MW three-blade horizontal-axis WT tower that is widely constructed on the southeastern coast of China was chosen for this study, as shown in Fig. 1(a). It was designed as Class IIa (a 10-min reference mean wind velocity of 42.5 m/s at hub height and a turbulence intensity of 0.16) in IEC 61400 (IEC 2005), without explicit considerations about its seismic performance. The tower is a near-cylindrical steel hollow tube. The outer diameter remains approximately 4,035 mm up to a height of 34 m and then reduces linearly from 4,035 to 2,955 mm for the remaining height. The shell thickness (t) varies from 25 to 10 mm, respectively. The position of the hub is 63.209 m above the base. The mass of the tower is approximately 91 t, and the rotor diameter is approximately 70 m. The mass of the blades and the nacelle located on top of the tower are 30 and 60 t, respectively. Generally, the WT is adjusted to the parked condition (i.e., the rotor rotational speed remains zero and the electric generator does not work) under extreme conditions using brake rigging to reduce potential rotor motions. Therefore, the parked condition is the scenario considered in this work.

The tower model was built in ABAQUS version 6.14-4, with the relevant details shown in Fig. 1(b). The tower wall was modeled by linear reduced-integration finite-strain S4R three-dimensional (3D) shell elements, which are suitable for dynamic and static analyses. The mesh in each wall segment was modeled using a tetrahedral sweep grid technique with increased resolution near both ends to better capture the increased failure vulnerability due to complex stress fields developed in these regions (Sadowski et al. 2017; Dai et al. 2017b). Based on the engineering drawings of the WT tower, an elliptical cut in the tower wall was introduced to consider the door at the bottom. This was reinforced in the perimeter by means of beam elements according to the actual design. Three internal stiffening flanges were included to increase the global bending

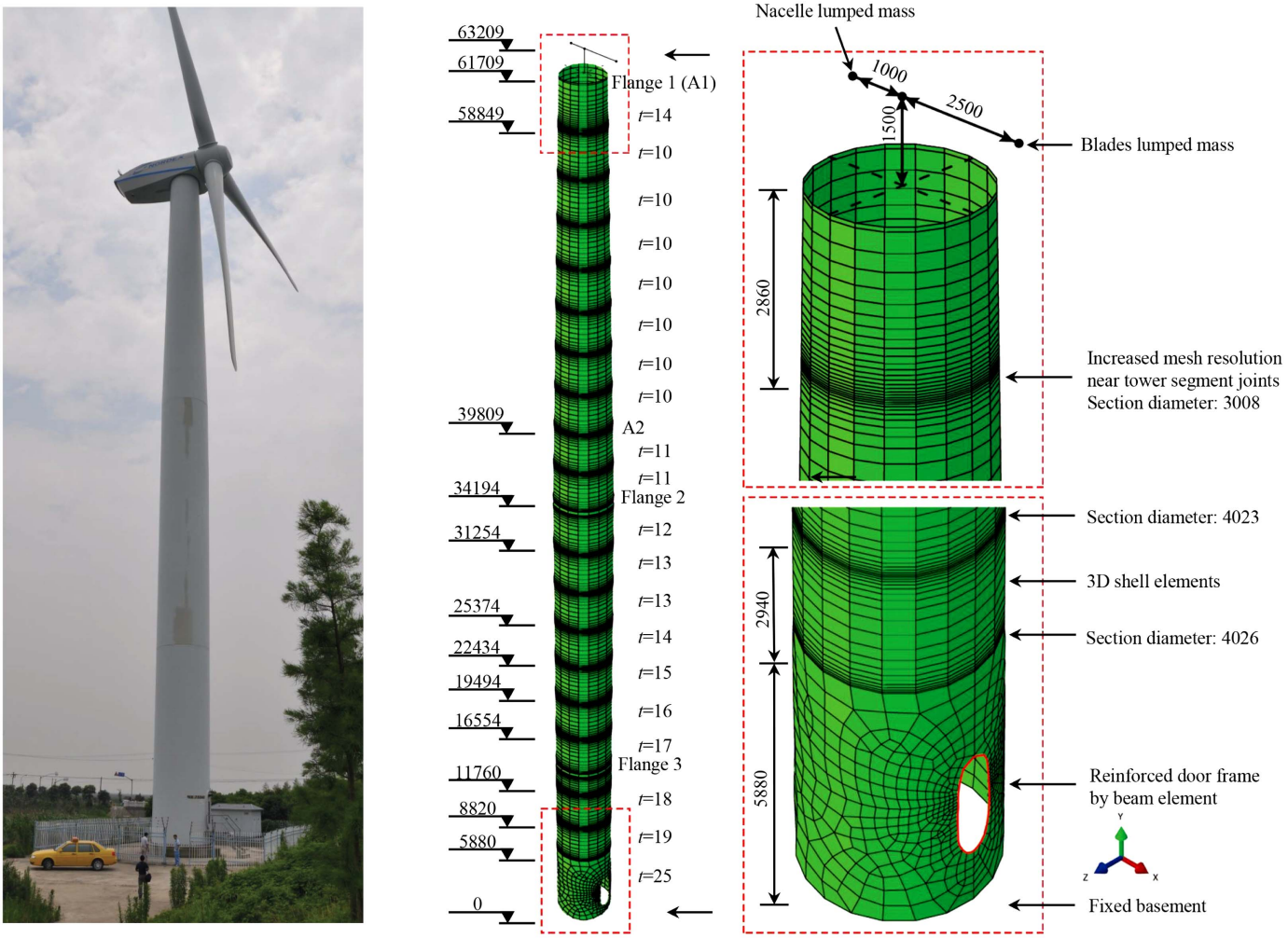


Fig. 1. 1.5-MW typical WT: (a) field study (image by Kaoshan Dai); and (b) details of FE model (unit: millimeters). "t" = wall thickness.

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Table 1. First three global flexural vibration modes of WT tower

Mode number	Direction	Calculated value (Hz)	Measured value (Hz)	Calculated value from Sadowski et al. (2017) (Hz)	Effective mass ratio (%)
1	X	0.49	0.49	0.48	69.79
	Z	0.49	0.48		69.71
2	X	4.32	3.84	4.17	14.33
	Z	4.42	4.08		14.53
3	X	12.02	—	—	5.80
	Z	12.84	—		6.64

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stiffness at the sections located 13.4, 34.2, and 61.8 m above the base. For simplicity, the stiffening flanges were modeled by 3D shell elements with the same dimensions as the actual flanges, ignoring screw and screw hole. The three flanges (1–3) from the top to the base have thicknesses of 80, 220, and 220 mm and extend 96.5, 133.5, and 137 mm from the tower wall, respectively. The nacelle and blades were simplified in the FE model as two lumped masses with kinematic coupling. This modeling approach was adopted in several works (Nuta et al. 2011; Patil et al. 2016; Sadowski et al. 2017; Dai et al. 2017b). Such a simplification is deemed appropriate because the focus is on the structural response of the tower in a parked condition. The foundation was considered completely rigid by constraining the displacements and the rotations at the bottom edge of the wall. Consequently, the soil–structure interaction is ignored.

The material in the wall is steel, and it is modeled as an ideal elastoplastic material with postyield strain hardening (0.1%). The steel type is S355 with a yield stress of 355 MPa, a Poisson's ratio of 0.3, an elastic modulus of 200 GPa, and a density of 7,850 kg/m^3. A self-contact rule along the entire tower was defined in the FE model to capture the potential contact between the surfaces of different wall segments under large deformation. Current research (Valamanesh and Myers 2014) suggests that a 1% damping ratio is reasonable in a 1.5-MW parked WT tower under earthquake actions. In this work, such structural damping was introduced by means of a Rayleigh damping based on the first and second vibration modes. This type of damping is justified because those two frequencies dominate the structural response, as will be demonstrated in this study.

Modal Analysis

The first three vibration modes related to the global flexure of the WT tower are presented in Table 1. The vibration modes in the two horizontal directions (X and Z) are very close to each other. The accuracy of the FE model to represent these vibration modes was supported by the good agreement between the numerical results and those obtained in a previous field dynamic measurement of the actual WT tower (Dai et al. 2015b, 2017c) as well as the numerical results from a previous study of the same WT (Sadowski et al. 2017), as shown in Table 1.

Dynamic Actions

Ground Motion Selection

Previous studies on the seismic response of WT tower classified earthquakes according to the design spectrum (Nuta et al. 2011; Asareh et al. 2016), fault distance, and pulselike effects (Patil et al. 2016; Sadowski et al. 2017). However, the frequency content of the ground motions was not directly considered in the selection process. Wind turbine towers usually have relatively low natural frequencies (around 0.5 Hz and below, as shown in Table 1), and the objective of this work was to study how earthquakes with different frequency contents may lead to different structural response and failure modes. The earthquake record selection was performed on the basis of the Pacific Earthquake Engineering Research database of strong ground motions (PEER 2013) to match the 5%-damping design response spectrum given in the Chinese seismic standard (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2010). The maximum spectral acceleration of this design response spectrum is $0.5g$, which was determined based on the specific site at which the field measurements were obtained for the proposed WT.

Two suites of ground motions with a different soil characteristic period (T_g) were selected. This period is defined as the one that marks the transition of the response spectra between the constant velocity and the constant acceleration regions. It is worth noting that the shear wave velocity was not a factor considered directly in the ground motion selection. A total of 20 ground motion records were selected, including 10 records with $T_g = 0.4$ s, which would correspond to stiff soil conditions (abbreviated as ST), and 10 records with $T_g = 1.1$ s, corresponding to soft soil (referred to as LT). Near-fault ground motions have distinct and potentially damaging characteristics such as forward directivity, fling-step, and hanging-wall effects (Yang and Wang 2012). Some of them show velocity pulses to which WT towers may be vulnerable (Stamatopoulos 2013; Patil et al. 2016). Therefore, all the ground motions considered in this work are near-fault, with a rupture distance below 20 km. Table 2 lists all the selected ground motion records, and Fig. 2 shows the target response acceleration spectrum and the spectra of each record in the two suites of ground motions. These spectra were obtained as the square root of the sum of the squares (SRSS) of the two horizontal components of each of the 10 ground motions in the set. Fig. 2 also includes the mean spectrum for each set, obtained as the arithmetic average of the 10 individual response spectra. The first two global flexural periods of the WT tower were included in the figure. Although the fundamental mode has a larger modal effective mass ratio than the second mode, as shown in Table 1, the associated spectral acceleration of the second mode is much higher, and this suggests that the structural response of the WT tower might be significantly influenced by the second mode as well. This will be discussed in the following section.

The ground motions were applied to the FE model of the WT as accelerations imposed at the tower base in the two horizontal directions (X and Z). The vertical component of the ground motion was not considered because of its moderate influence on the response of the WT tower (Nuta et al. 2011; Sadowski et al. 2017). To reflect the elastoplastic performance of the WT tower, the peak ground accelerations (PGAs) of all the ground motions were scaled to four levels: $0.1g$, $1g$, $2g$, and $3g$.

Table 2. Ground motion records and scale factors applied to match target spectrum

Number	Matching T_g (s)	Earthquake name	Year	Station	V_{s30} (m · s^{-1})	Scale factor
1	0.4	Imperial Valley-06	1979	El Centro Array Number 3	162.94	0.52
2	0.4	Chi-Chi	1999	TCU049	487.27	0.55
3	0.4	Imperial Valley-06	1979	Holtville Post Office	202.89	0.52
4	0.4	Imperial Valley-06	1979	Brawley Airport	208.71	0.69
5	0.4	Darfield	2010	DSLC	295.74	0.54
6	0.4	Superstition Hills-02	1987	Brawley Airport	208.71	1.31
7	0.4	Imperial Valley-06	1979	Parachute Test Site	348.69	1.17
8	0.4	Chi-Chi	1999	TCU122	475.46	0.56
9	0.4	El Mayor-Cucapah	2010	Chihuahua	242.05	0.58
10	0.4	Iwate	2008	Mizusawaku Interior	413.04	0.52
11	1.1	Chi-Chi	1999	TCU065	305.85	0.36
12	1.1	Imperial Valley-06	1979	EC County Center FF	192.05	0.77
13	1.1	Darfield	2010	GDLC	344.02	0.33
14	1.1	Northridge-01	1994	Sylmar—Converter	251.24	0.29
15	1.1	Superstition Hills-02	1987	Parachute Test Site	348.69	0.44
16	1.1	Chi-Chi	1999	TCU120	459.34	0.80
17	1.1	Cape Mendocino	1992	Fortuna—Fortuna Blvd	457.06	1.46
18	1.1	Duzce	1999	Lamont 1058	529.18	2.36
19	1.1	Darfield	2010	LRSC	295.74	2.06
20	1.1	El Mayor-Cucapah	2010	Cerro Prieto	242.05	0.61

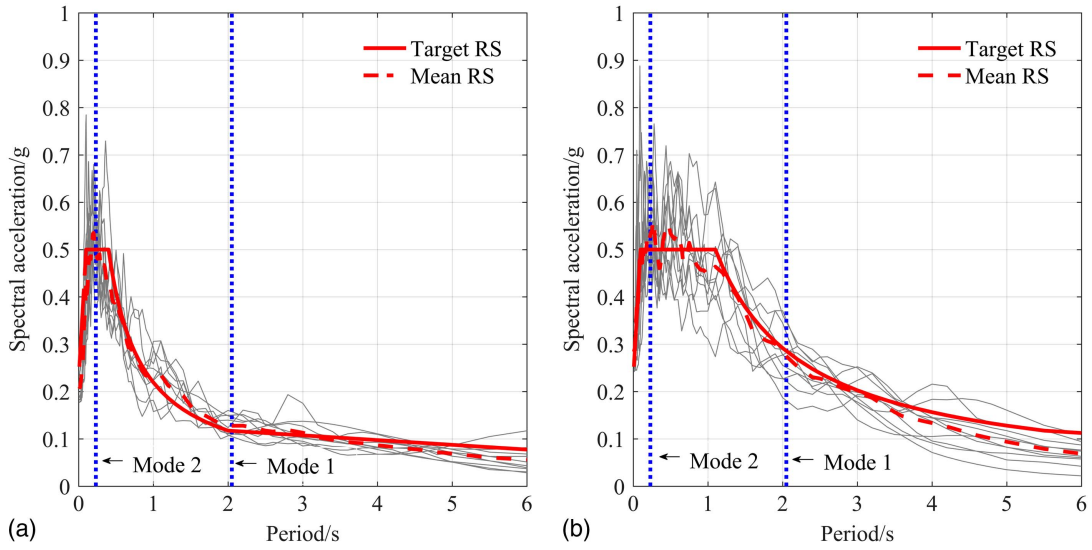


Fig. 2. Ground motion acceleration response spectra and first two vibration periods of structure: (a) ST records, $T_g = 0.4$ s; and (b) LT records, $T_g = 1.1$ s.

Wind Load Generation

The strong wind velocity field was simulated using the consistent discrete random flow generation (CDRFG) method (Aboshosha et al. 2015). This made it possible to generate wind velocity time histories at different points of the WT tower that are consistent with a target wind velocity power spectrum while respecting the corresponding spatial and temporal correlations. Three levels of the mean wind velocity at the hub level are considered in order to reach the nonlinear response and potentially the collapse of the WT tower, namely, $U_{ref} = 50$, 55, or 60 m/s (referred to as Cases 1–3). The generation of the wind was based on the sum of harmonics, with a 0.2-Hz frequency interval and lower and upper cut-off frequencies of 0.01 and 10 Hz, respectively. The duration of the generated signals was 600 s, and the time interval was 0.1 s. The remaining parameters relevant to the simulation of the wind field are listed in Table 3, and more details can be found in Dai et al. (2017b). A total of three independent wind velocity fields were generated for each of the three values of the reference 10-min mean wind velocity to account for the record-to-record variability in the structural response.

Blade element momentum (BEM) theory (Hansen 2008) was used to obtain the wind load on the blades. Each blade was discretized with 17 elements along its length. The lift and the drag forces depend on the instantaneous angle of attack of the wind. Moreover, the thrust force and the tangential force induced on each blade element were calculated. The total forces on the blades applied to the tower top result from the summation of these element forces.

The wind load time histories on the tower body and the nacelle were obtained as

$$F(t) = 1/2C_d\rho U^2(t)A \quad (1)$$

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Table 3. Parameters used in wind velocity generation

Parameter	Used model	Model parameter description and value
Mean velocity profile	$U(z) = U_{ref}(z/Z_{ref})^{\eta}$	z = height Power exponent $\eta = 0.16$ [rural area defined in Ministry of Housing and Urban-Rural Development of the People's Republic of China (2012)] Hub height $Z_{ref} = 65$ m Hub mean velocity $U_{ref} = 50$, 55, or 60 m/s
Turbulence intensity profile	$I(z) = I_{ref}(z/Z_{ref})^{-d}$	Power exponent $d = 0.18$ Hub height turbulence intensity $I_{ref} = 0.16$ [Class A in IEC 61400 (IEC 2005)]
Turbulence length scale profile	$L_u(z) = L_{u(ref)}(z/z_{ref})^{\gamma}$	Hub height turbulence length scale $L_{u(ref)} = 77$ m Reference height $z_{ref} = 10$ m Power exponent $\gamma = 0.3$
Turbulence power spectrum	$\frac{fS(f)}{\sigma_u^2} = \frac{4(fL_u(z)/U(z))}{(1+70.8(fL_u(z)/U(z)^2)^{5/6}}$ (von Karman model)	σ_u^2 = standard deviation of fluctuating velocity f = frequency
Coherency function	$Coh(f_m) = \exp(-C_j f dx_j/U(z))$	Coherency decay constant $C_j = 10$ dx_j = distance between correlative points

where the air density $\rho = 1.25$ kg/m^3; $U(t)$ = instantaneous wind velocity time history; C_d = drag coefficient, which is 1.2 for the WT tower and 1.3 for the nacelle; and A = projected area.

In the event of extreme winds such as tropical cyclones, the WT is generally parked and the blades are in a "feather" state to reduce the wind load in the fore–aft direction. However, due to the rapid changes in the wind direction that characterize strong wind events, the inflow wind may be parallel to the side-to-side direction (i.e., the Z direction in the FE model) of the WT tower, which was shown to be critical for the response of the WT (Wang et al. 2013; Dai et al. 2017b). Consequently, the wind inflow in this work was considered to be in the side-to-side direction.

Failure Analysis and Discussion

Elastic Structural Response

The study of the structural response started by maintaining the tower in the elastic analysis range in order to have a clear understanding of the participation of different vibration modes. To this end, selected ground motions scaled to 0.1g PGA were considered, for which the WT tower remains completely elastic for all records. The time-history responses of the relative horizontal displacement at the top of the tower and the bending moment at the base were extracted from the FE model. The time-history results in the two horizontal directions were combined for each earthquake using the SRSS method, and the arithmetic mean, the root-mean square (RMS), and the peak response for the ST and LT ground motions were obtained (Table 4).

The results indicate that the seismic response of the WT tower in an elastic range is the largest under LT ground motions. This can be explained by the higher spectral acceleration associated with the fundamental vibration period $[S_a(T_1)]$ for LT earthquakes in comparison with ST records under the same PGA, as shown in Fig. 2.

In addition, the accelerations recorded at the tower sections for which the modal displacement in the first two vibration modes is the maximum were studied for two representative ST and LT earthquakes, namely, Records 4 and 17 in Table 2, respectively. These two sections correspond to Points A1 and A2 in Fig. 1, at heights of 61.8 and 39.8 m from the tower base, respectively. The accelerations at these sections were also obtained under wind excitations with a mean velocity of 50 m/s at the hub height (Case 1), for which the structural response is also elastic. The two horizontal (X and Z) components of the acceleration signals were processed to obtain their power spectral density (PSD)s, which were combined with the SRSS method and normalized with respect to the peak value at the fundamental frequency. The normalized PSD curves and the mode shapes of the first two vibration modes were included in Fig. 3 for tower Sections A1 and A2, respectively. Fig. 3(a) shows that the response of the tower top (A1) is clearly dominated by the fundamental mode. However, Fig. 3(b) illustrates the important contribution from the second vibration mode (4.32–4.42 Hz) to the acceleration at the intermediate section of the tower (A2) under seismic actions. This is especially significant for the ST earthquake, for which the second mode dominates the response at the intermediate region of the tower. This is due to the relatively large spectral acceleration associated with the second vibration mode for ST ground motions, in comparison to that in the first mode [Fig. 2(a)], and the large participation (modal displacement) of the second mode at this part of the tower. The results also confirm that the elastic response of the upper half of the WT tower can be captured with the first two vibration modes. Regardless of the position of the section in the upper half of the tower, the response is always dominated by the fundamental mode when the structure is subjected to wind loading. This is explained by the relatively large energy of the wind actions in the low-frequency region.

Table 4. WT tower seismic response with PGA of 0.1g

Ground motion type	Top displacement (m)			Base moment (kN · m)		
	Mean	RMS	Maximum	Mean	RMS	Maximum
ST	0.038	0.047	0.12	2,538	3,181	8,503
LT	0.065	0.082	0.19	4,338	5,472	13,119

Note: LT = long vibration period; RMS = root-mean square; and ST = short vibration period.

Representative Failure Modes under Strong Seismic Actions or Wind Loads

A series of nonlinear dynamic FE analyses were conducted for the 20 seismic records under 3 PGA levels (1g, 2g, and 3g) and 3 different wind velocity fields generated based on 3 intensity levels (50, 55, and 60 m/s). Hilber-Hughes-Taylor time integration (Hilber et al. 1977) with automatic step time was used as solver for

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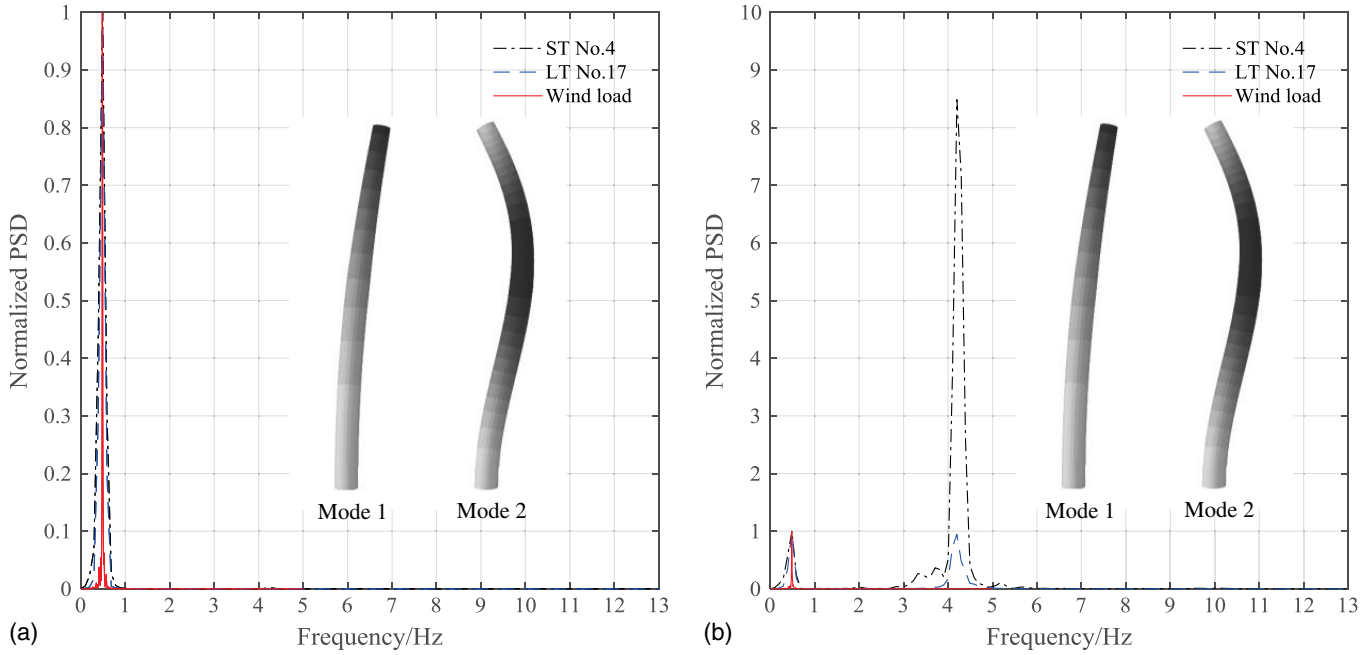


Fig. 3. Acceleration PSD at Sections (a) A1 and (b) A2 under earthquake (PGA $= 0.1g$) or wind loads ($U_{ref} = 50$ m/s).

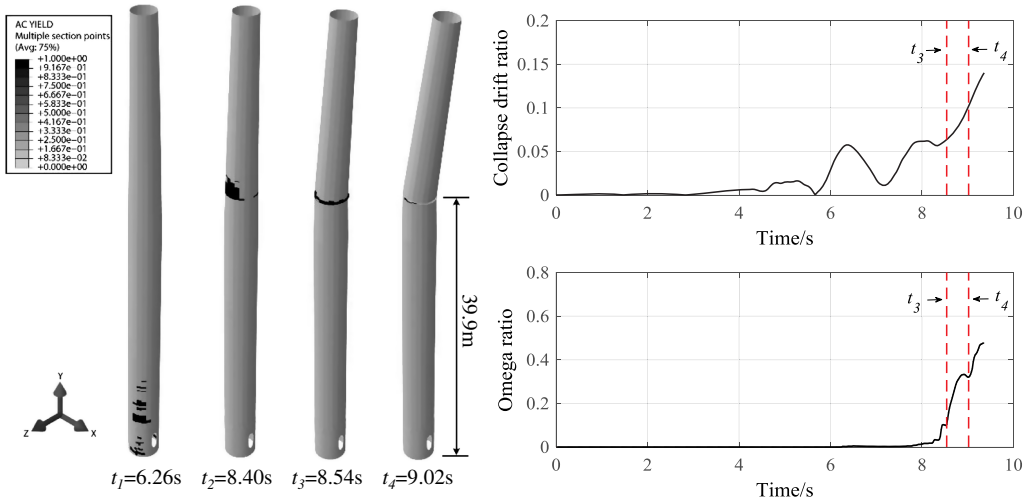


Fig. 4. Failure mode for No. 3 ST ground motion; PGA $= 2g$.

nonlinear time-history calculation. Four different failure modes were observed (Figs. 4–7) among these nonlinear dynamic responses analyses. Additionally, four stages were identified in the failure process: Stage 1, initial plastic hinge formation; Stage 2, plastic hinge development; Stage 3, formation of full-section plastic hinge; and Stage 4, local buckling. The normalized yielding contour map of the tower for each stage is included in Figs. 4–7 to illustrate the failure process, and the four stages were identified at time intervals of t_1, t_2, t_3, and t_4. In this contour map the value 1 (black color) represents yielding.

The progressive failure of the WT tower under seismic or wind actions shows that the plastic hinge begins at the base of the tower. However, as the input excitation energy increases, the damage spreads toward the top of the tower, and the yielding of the steel tends to be concentrated at the welding between the different rings composing the tower wall, where there is a change in thickness. This change in thickness leads to local stress concentration, which can result in the formation of plastic hinges. Collapse is eventually reached when a single section of the WT tower yields along its entire perimeter, forming a full plastic hinge. At this moment the structure becomes a mechanism in which the part of the WT tower above the plastic hinge rotates almost freely. Under both load types, the tower usually collapses in a short time once the full-section plastic hinge occurs due to the low structural redundancy of the WT tower.

The position of the full plastic hinge WT tower was observed to occur at four different locations under different strong ground motions. However, when the WT tower is subject to extreme wind loads, the failure is always concentrated at a single location that is positioned 8.8 m above the base of the tower. Due to the variation of the failure location under different actions, the traditional damage index based on the residual deformation considering the full

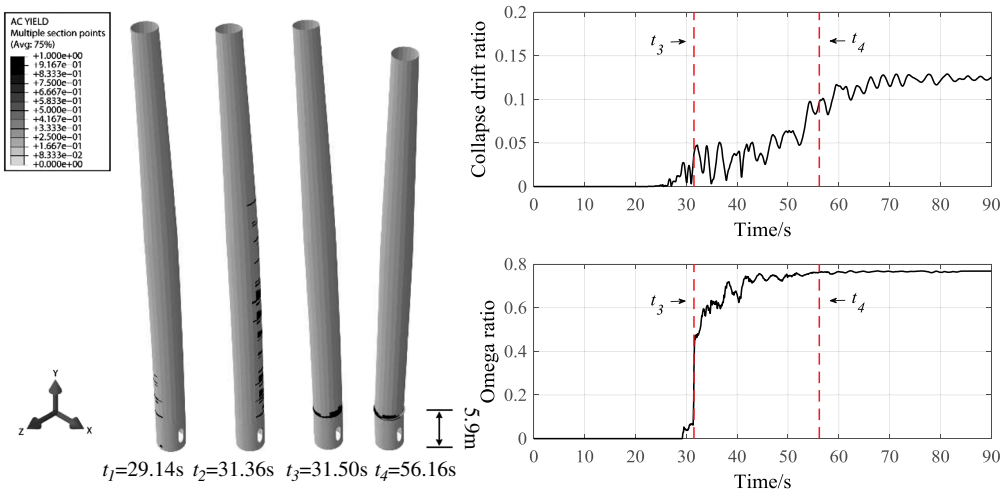


Fig. 5. Failure mode for the No. 8 ST ground motion; PGA $= 3g$.

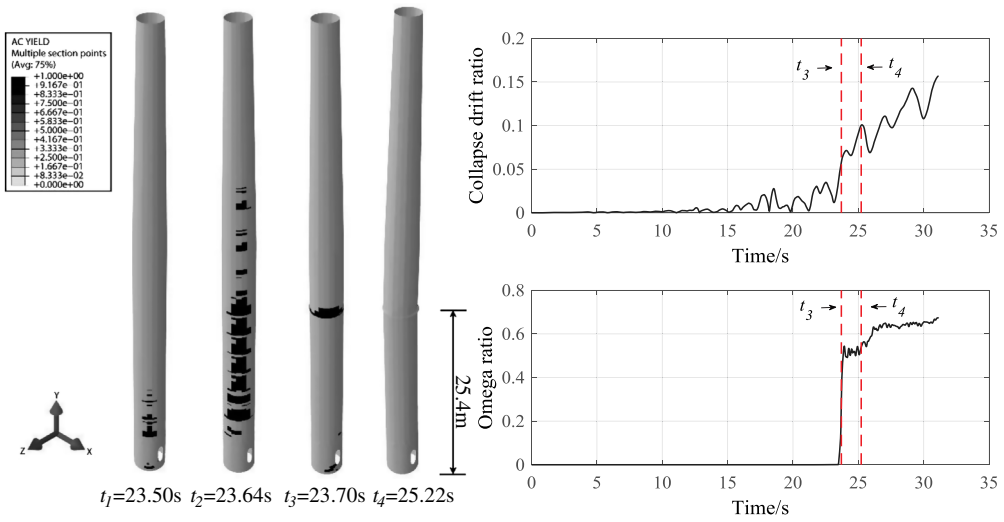


Fig. 6. Failure mode for the No. 19 LT ground motion; PGA $= 3g$.

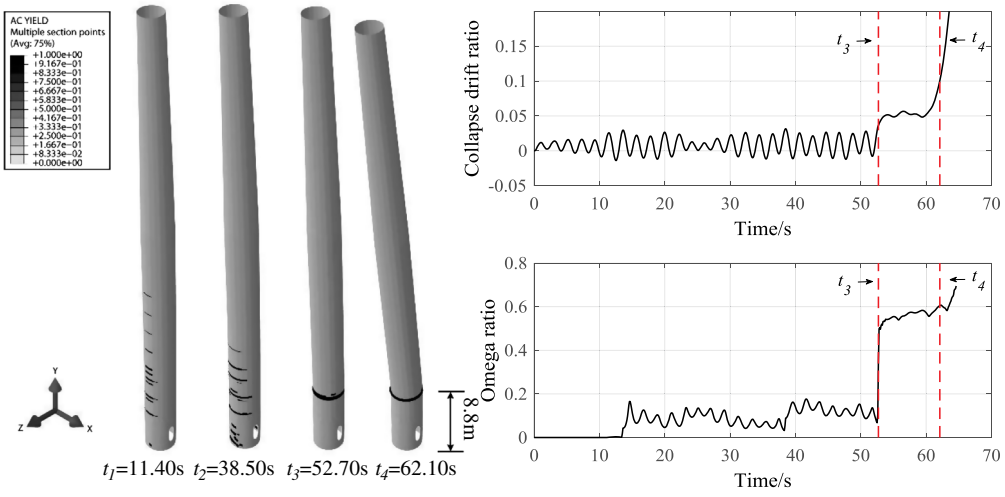


Fig. 7. Failure mode for No. 1 wind record; $U_{ref} = 55$ m/s (Case 2).

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tower height (Nuta et al. 2011; Asareh et al. 2016) was modified. The following collapse drift ratio (CDR) is proposed to consider the effect of different failure heights:

$$CDR = D_t/(H_t - H_c) \quad (2)$$

where D_t = displacement at tower top; H_t = tower height; and H_c = height at failure. In all the cases, the height is measured from the base of the tower.

In addition, the Omega ratio proposed by Camara et al. (2017) was calculated to study the energy dissipated by plasticity at different locations of the tower during the analysis. The Omega ratio was defined as the cumulative dissipated plastic energy divided by the total external work at the instant of the analysis for which it is calculated. Both energy quantities were defined in the entire tower. The evolution of the CDR and the Omega ratio for the whole tower is shown in Figs. 4–7. It is clear that both indexes increase significantly during the development of the plastic hinge (Stage 2). Two characteristic values of the time-history response are marked with vertical dashed lines in the figures, and these values were later processed in the failure mode statistics: (1) the time instant when the full-section plastic hinge is formed, t_3, and (2) the time instant when the CDR reaches 0.1, t_4. It is observed that, in general, the lower the position of the full plastic hinge, the higher the plastic energy absorbed by steel hysteresis of the tower wall and, therefore, the larger the Omega ratio. In Fig. 4, since the position of the plastic hinge is relatively high, the capacity of the structure to absorb the seismic energy is limited and the Omega ratios are 10% and 32% for the aforementioned two time instants, respectively. This is regarded as a brittle and undesirable response because the structure exhibits a low capacity to dissipate the seismic energy before the collapse occurs. The failure mode for the plastic hinge at the top of the tower in Fig. 5 is in clear contrast with that observed in Fig. 4, in which the plastic hinge is located at the bottom part of the tower (5.9 m above the base), and because of this, up to 76% of the seismic energy is absorbed by plasticity before collapse. The larger dissipation capacity of the bottom sections of the tower is due to the larger radius and thickness of the wall in this region.

From the perspective of collapse duration ($t_4 - t_3$), middle-upper failure tends to have associated responses with a relatively low dissipation capacity and a fast development of the full plastic hinge (only 0.5 s from initiation to CDR = 0.1 in Fig. 4). This is in contrast to those failure cases where full plastic hinges occur at the bottom part of the tower (25 and 10 s from damage initiation to CDR = 0.1 in Figs. 5 and 7, respectively).

Failure Mode Statistics

The tower failure locations for the nonlinear time-history seismic and wind analyses are summarized in Tables 5 and 6, respectively. In a few cases, denoted by the symbol "+" in the tables, the tower enters failure Stages 1 and 2 but does not Stage 3. The symbol "–" indicates that the tower remains in an elastic range for the entire analysis. The results show that the LT ground motions are more prone to induce localized yielding (Stages 1 and 2) and the development of a full-section plastic hinge with buckling failure (Stages 3 and 4), which is on account of the higher spectral acceleration associated with the fundamental mode [$S_a(T_1)$] in LT ground motions, as discussed previously.

All the local buckling failure cases given in Tables 5 and 6 and illustrated in Figs. 4–7 for particular records occur at locations where the tower wall thickness changes, namely, at 5.9 m above the base (9% of total height, the thickness changes from 25 to 19 mm), 8.8 m (14% height, the thickness changes from 19 to 18 mm), 25.4 m (39% height, the thickness changes from 14 to 13 mm), and 39.9 m (61% height, the thickness changes from 11 to 10 mm). The results indicate that the change in thickness concentrates the stresses of the wall in bending under lateral dynamic actions and leads to the potential development of a full plastic hinge.

Table 5. Failure locations induced by ground motions

Number	Ground motion type	Failure height for PGA = $1g$	Failure height for PGA = $2g$ (m)	Failure height for PGA = $3g$ (m)
1	ST	–	5.9	5.9
2	ST	–	+	5.9
3	ST	–	39.9	39.9
4	ST	–	5.9	5.9
5	ST	–	+	25.4
6	ST	–	+	5.9
7	ST	–	–	39.9
8	ST	+	25.4	5.9
9	ST	+	25.4	39.9
10	ST	–	+	39.9
11	LT	+	25.4	25.4
12	LT	–	5.9	5.9
13	LT	+	8.8	39.9
14	LT	–	8.8	8.8
15	LT	8.8	25.4	39.9
16	LT	8.8	5.9	25.4
17	LT	8.8	39.9	5.9
18	LT	–	8.8	25.4
19	LT	–	25.4	25.4
20	LT	8.8	25.4	39.9

Note: LT = long vibration period; ST = short vibration period; + = tower enters failure Stages 1 and 2 but does not reach failure Stage 3; and – = tower remains in elastic range during analysis.

Table 6. Failure locations induced by strong winds

Number	Failure height for U_{ref} = 50 m/s (m)	Failure height for U_{ref} = 55 m/s (m)	Failure height for U_{ref} = 60 m/s (m)
1	–	8.8	8.8
2	–	8.8	8.8
3	+	8.8	8.8

Note: + = tower enters failure Stages 1 and 2 but does not reach failure Stage 3; and – = tower remains in elastic range during analysis.

The comparison of the Omega ratio for all failure cases with the full-section plastic hinge and with CDR = 0.1 is shown in Fig. 8. The ordinates represent the failure height and the abscissas the corresponding Omega ratio. The results are consistent with those discussed in section "Representative Failure Modes under Strong Seismic Actions or Wind Loads," with the exception of the failure at a height of 5.9 m. The mean Omega ratio at this location is lower than that for failure locations at 8.8 and 25.4 m, which may result from the fact that the change in thickness at 5.9 m is significant (6 mm). But the general trend is that the percentage of the energy dissipated by plasticity (Omega ratio) is reduced with the increase in failure height.

From Tables 5 and 6 it can be seen that the collapse usually occurs at sections that are located in the lower half of the tower. More failures were observed at the middle-upper regions (39.9 m) for the ST ground motion cases (33%) than for the LT ones (17%). Since a larger Omega ratio can be reached as the plastic hinge is closer to the base of the tower, the wall tends to dissipate more seismic energy under the LT ground motions than under the ST motions. In contrast to seismic loading, the WT tower always failed at the bottom location under strong winds, which is 8.8 m above the

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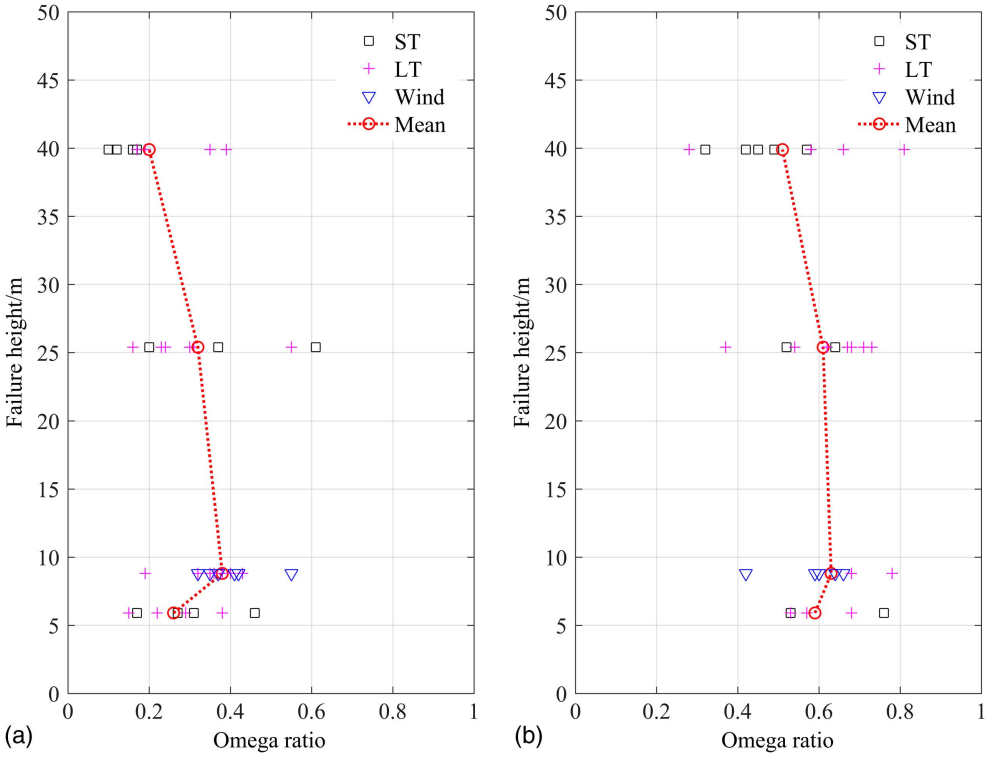


Fig. 8. Omega ratio at (a) initation of full-section plastic hinge; and (b) $CDR = 0.1$.

tower base. Accordingly, relatively large plastic dissipation was observed under the wind actions.

Pushover Analysis

The variation in the failure modes under different dynamic actions is attributed to changes in the participation of the vibration modes under loads with different frequency contents. To explore this effect further, nonlinear static (pushover) analyses were conducted. Most of the previous pushover analyses of WT towers considered exclusively the contribution of the fundamental mode to obtain the load pattern (Patil et al. 2016) or directly applied a single point load or displacement at the tower top (Guo et al. 2011; Asareh et al. 2016). However, the contribution of higher-order modes can be important in seismic responses, as indicated from the foregoing nonlinear time-history analyses, especially for the cases under ST ground motions. Modal pushover analyses based on the inertia forces associated with the first two modes were conducted. For each vibration mode, the load at the ith element of the FE model was obtained as

$$F_i = m_i \phi_i \tag{3}$$

where m_i and ϕ_i = mass and modal displacement in ith element, respectively.

In addition to the modal pushover that was performed to explore the seismic response, the mean wind load pattern along the height of the tower was considered to study the WT failure under the wind load. The mean wind velocity profile and the approach to obtain wind loads were described in section "Wind Load Generation."

The loads in the static pushover analyses increased progressively until the tower failed. This stage is illustrated in Fig. 9. In those cases corresponding to the first and second mode load patterns, the structure fails at the cross sections that are 8.8 and 42.8 m above the ground, respectively. The failure modes observed from the pushover analyses are basically consistent with the time-history analysis. It is evident that the failure location of the tower moves upward under the second mode load pattern. This effect is consistent with observations in the dynamic analysis when the contribution of the second mode increases, which is more likely to occur under ST earthquakes, as discussed previously. Under the wind load induced by the mean velocity profile, the WT tower collapses due to the formation of a plastic hinge at the bottom part of the structure (5.9 m from the base).

Frequency Content of Response under Earthquake and Wind Actions

Fig. 10 compares the PSDs of three accelerograms (ST No. 3, LT No. 14, LT No. 19) normalized with respect to the peak value at the fundamental frequency. The reason for selecting these particular records is that they trigger different failure locations in the tower. A normalized PSD of a wind velocity time history at the hub height ($U_{ref} = 50$ m/s, Record No. 1) was also included for comparison purposes. It can be seen that the frequency content of the earthquakes is characterized by a broad range of frequencies, which contributes to activating high-order modes. On the other hand, the wind action tends to have more energy associated with the fundamental frequency of the tower, which causes it to dominate the global response.

It is again observed that the frequency content of the lateral loads has significant effects on the tower collapse modes. The LT ground motion Record No. 14 is dominated by low frequencies, resulting in the failure at the bottom part of the tower. However, ST Record No. 3 has a large value of the PSD in the frequency range in which the second structural mode is included, which leads to the failure of the tower at the middle-upper region of the structure (Fig. 4). LT Record No. 19 is richer in a broader range of frequencies compared to Record Nos. 14 and 3, and the failure for this ground motion occurs at the middle-low region of the tower (Fig. 6).

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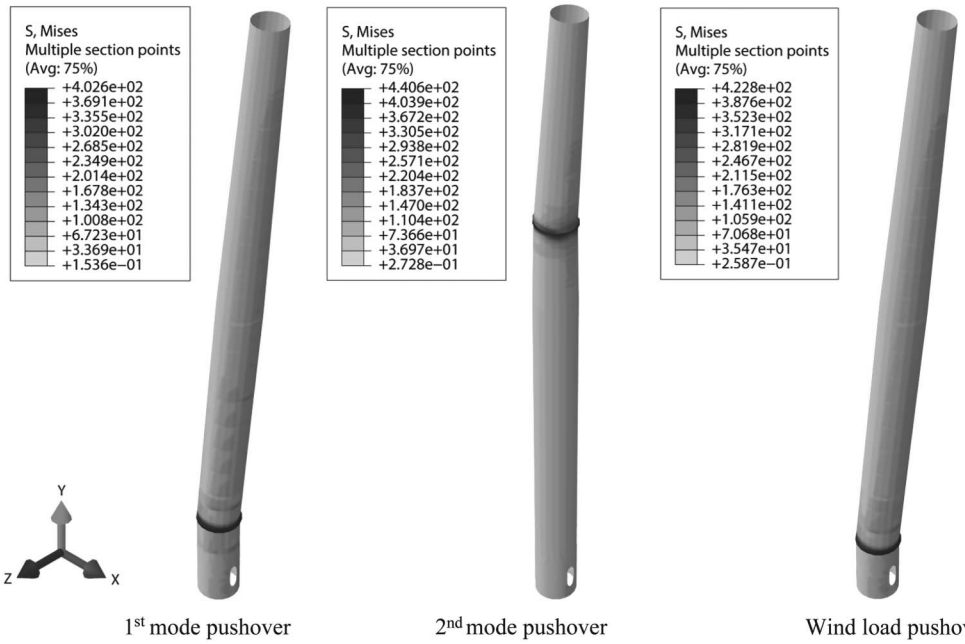


Fig. 9. WT failures under loads associated with first two modes and wind load induced by mean velocity profile. The von Mises stress is included in the contour plots, with units in MPa.

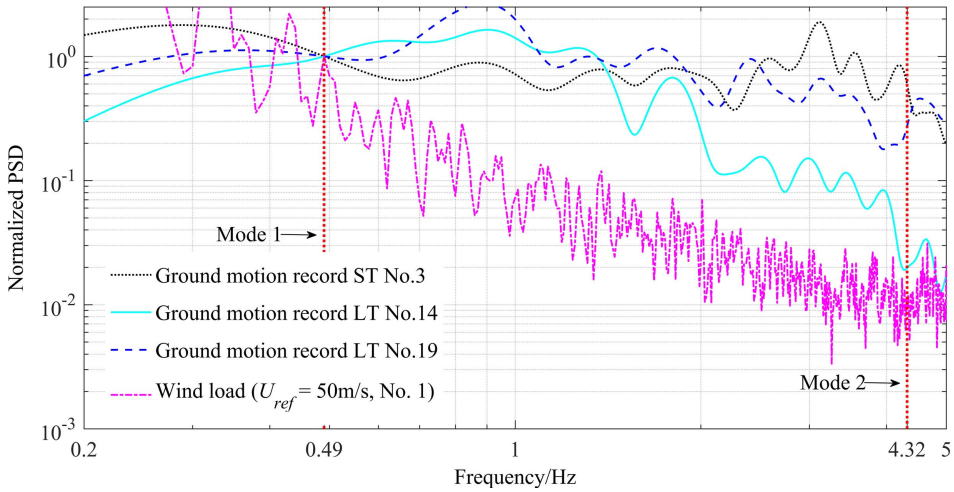


Fig. 10. Normalized PSDs of three ground motion time histories and one wind record.

In general, for ST ground motions the PSD verifies that the participation of the second tower vibration mode [Figs. 2(a) and 10] is more significant compared to LT earthquakes and wind excitations, which then tends to shift the failure location of the tower upward. It should be acknowledged that, due to the record-to-record variability of the ground motions, the failure location may change under different excitations. However, the progressive collapse mode associated with the frequency content of external loads provides general information for the design of WT towers in wind farms located in hazardous regions.

Conclusions and Recommendations

A typical wind turbine under extreme loads was studied in this paper based on a detailed finite-element model of the tower. Two suites of ground motions with different frequency content features were selected, and a series of wind velocity time histories was generated. A total of 89 nonlinear dynamic analyses were performed, and additional nonlinear static (pushover) analyses were conducted. The results led to the following findings:

1. The elastic response of the WT tower is larger under earthquakes dominated by long vibration periods (LT) than for those with a significant short period (ST) frequency content. This is attributed to different participations of the tower vibration modes in the structural response.
2. Under lateral loads, yielding initiates at the bottom part of the tower wall. As the loads increase, the damage is concentrated in the welding connections between different rings in the tower, where the thickness of the wall changes. Once the entire perimeter of a section yields, a full plastic hinge is developed and the tower collapses rapidly due to its low structural redundancy.

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3. The locations of the full-section plastic hinge depend on the frequency content of the excitation. The wind loading mainly excites the fundamental mode of the structure, resulting in the failure at the bottom half of the tower. This is also the case for most LT ground motions. ST earthquakes, however, are more likely to trigger the failure at the middle or upper part of the tower, owing to the influence of the second vibration mode. A relatively low dissipation capacity and rapid collapse is observed when the plastic hinge is formed in the upper part of the tower.
4. Pushover analysis can provide useful insight into the potential failure modes of the tower under dynamic excitations. It is recommended that load patterns associated with high-order vibration modes be considered in order to capture dangerous failure cases, in which failure occurs in the middle or upper regions of the tower.

The following recommendations can be made for improving WT performance under these types of extreme loading in practice:

1. Special attention should be paid to the design of WTs to be located in regions where the seismic risk may motivate higher-mode responses (e.g., WT around 60 m in height under ST ground motions). It is recommended that the walls be relatively thicker at the middle and upper parts in order to facilitate the development of the plastic hinge as close to the foundation as possible, which should be able to sufficiently dissipate energy.
2. For strong wind conditions that mainly motivate the fundamental mode of the WT, the bottom part of the structure is more vulnerable. It is recommended that the bottom half of WTs located at sites dominated by strong wind loadings be reinforced.
3. Sharp changes in thickness should be avoided in the WT tower to maximize plastic energy dissipation and minimize the risk of sudden collapse.

Since the preceding findings are based on the analysis of one representative WT 63.2 m in height, to extend the results, more investigations are still required for other WTs with different dynamic characteristics. Additionally, the nonlinear response considering soil–structure interaction and aerodynamic and seismic coupling effect should be further studied to make the conclusions more comprehensive.

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